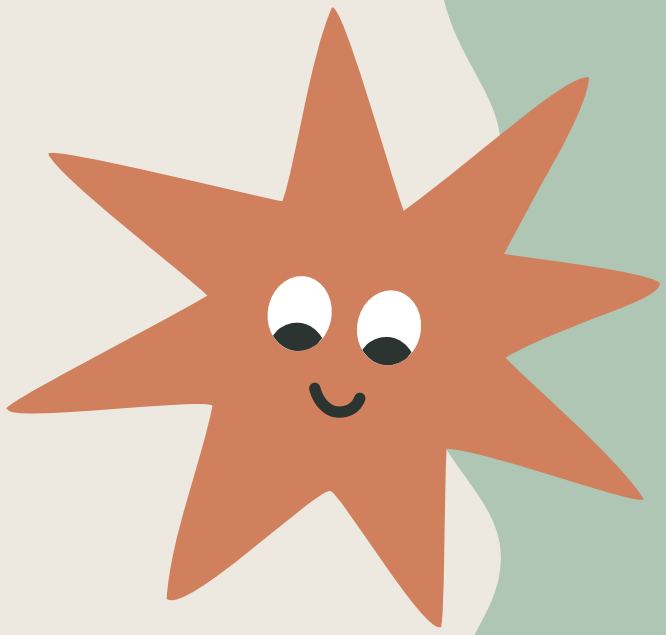
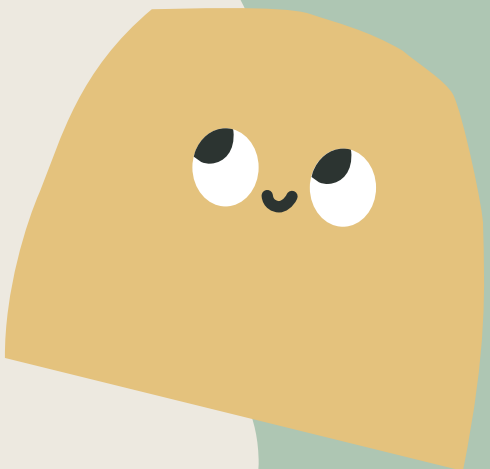
Artificial Intelligence Use in Statistical Programming

BST 430

TJ Weaver





Agenda

- Tools Available
- **Code / Software Projects (FLEX)**
- Conceptual Understanding
- Academic Writing
- Mathematical Work
- Presentations
- Conclusion / Thoughts on Future

Tools Available

Models

UR AI

- Built on Open AI's GPT-4 model
- **Approved tool for use with high-risk / non-public University of Rochester data**
- Ensures privacy and compliance with University of Rochester policy
- Accessible at **chat.rochester.edu** and sign in with your university email and password
- No usage limits apply

CLAUDE

- Commonly used in industry w/ corporate subscription (individual pro subscription cost $17 / month)
- Can be implemented to read & write code directly within a codebase or github repository on a private device
- Good for complex tasks
- **Not approved for use with high risk / non-public University of Rochester data**
- Various levels of effort available, usage limits usually apply based on tokens

Claude Code



- Included with a pro subscription at no extra cost
- Open PowerShell/Termincal and run “irm https://claude.ai/install.ps1 | iex”
- Close and reopen PowerShell/Terminal and run “claude –version”
- Use cd to navigate to your desired project folder
- Run “claude” and you should see a login prompt and option to open your browser, open your browser and log in with your pro-subscribed account (may need to press approve/activate after log in)
- Navigate back to Terminal/Powershell when you see some kind of “all set” on your browser
- Claude should be working / open and ready on Terminal or Powershell
- Whenever you’d like to use Claude Code, simply navigate to the desired folder and in Terminal / Powershell and type “claude”
- **Do not open claude code in folders containing high risk or unauthorized University of Rochester data**

Other Common Models

- Chat GPT (OpenAI)
- DeepSeek
- Gemini (Google)
- Grok (X)
- Llama (Meta)



Code / Software Projects (FLEX)

FLEX – Framing, Logistics, Execution, *e*Xplanation

- This is basically just my own ideas about how to ensure AI remains within bounds on complex software projects
- It is important to break up complicated tasks into a number of manageable pieces; this decreases the likelihood of mistakes and increases the chance of catching errors, flaws in reasoning, or deviations from your intentions
- **F**RAMING: Write out **context, goals, and reasoning** behind the project and the proposed code – this makes all subsequent steps easier for AI, and increases the likelihood that it will ask questions where it is unsure and can suggest improvements
  LOGISTICS: Explain that the task needs to be completed in several steps and that you will ask it to code each step separately, then **lay out all steps**
- Ask AI to complete the first step, adding any additional necessary details
- **E**XECUTION: Verify that the first step runs as intended and outputs are sensible
- e**X**PLANATION: Read all code and comments, ensure that you understand the logic and that inputs and outputs are as you expect – conduct sanity checks (eg if AI writes a function try it for a simple case to ensure it is working as expected), make your own comments as needed
- Ask AI to complete the second step, execute, explain, and repeat for all steps – at each step add as much detail as is necessary for completion of the step

FLEX – Example

- FRAMING: Hi agent, I am currently working with the famous iris dataset in R. The data contains information about three different species of flowers, but there is genuine debate in the scientific community about whether two of these species genuinely differ, so I would like to use some statistical methods as well as some visuals to help me determine my opinion.
- LOGISTICS: First, I would like to create several scatter plots with ggplot for visual comparison of the species. Second, I would like to run two statistical methods to see how the methods cluster, and plot scatter plots of their clustering as well. I want to plot fitted clustering with colors and “true” clusterings with shapes.
- EXECUTION: Now do the first step: write code to load in the iris data and plot ggplot scatter plots of sepal.width against sepal.height, where species 1 is circles, species 2 is triangles, and species 3 is squares. Thanks!
- AI output (eXPLAIN by reading yourself and ensuring all outputs are sensible):

Sure. One small correction: in the built-in iris dataset, the variables are Sepal.Width and Sepal.Length—there is no Sepal.Height. Assuming by height you mean sepal length, you can do:

```r
library(ggplot2)

# Load the built-in iris dataset
data(iris)

# Look at the species labels
levels(iris$Species)
```

```r
# Scatter plot: Sepal Width vs. Sepal Length
ggplot(iris, aes(
  x = Sepal.Width,
  y = Sepal.Length,
  shape = Species
)) +
  geom_point(size = 3) +
  scale_shape_manual(
    values = c(
      "setosa" = 16,      # circle
```

```r
      "versicolor" = 17,  # triangle
      "virginica" = 15    # square
    )
  ) +
  labs(
    x = "Sepal Width",
    y = "Sepal Length",
    shape = "Species",
    title = "Iris Species: Sepal Width vs. Sepal Length"
  ) +
  theme_minimal()
```

FLEX – Example (Continued)

- EXECUTION (step 2 of planned analysis): Great! Can you now write a chunk that uses k means with k = 2 on this data to cluster the flowers, and then plots the ggplot with color used to separate by cluster and shape used to separate by true flower type?
- AI output (eXPLAIN by reading and understanding):

This chunk runs **k-means with (k=2)** using all four numeric flower measurements, adds the fitted cluster assignment to iris, and then plots the same sepal-width/sepal-length scatter plot with **color = k-means cluster** and **shape = true species**.

This should be particularly useful for your question because the **colors show what a completely unsupervised two-group solution thinks**, while the shapes let you see where those two fitted clusters cut across the three known species.

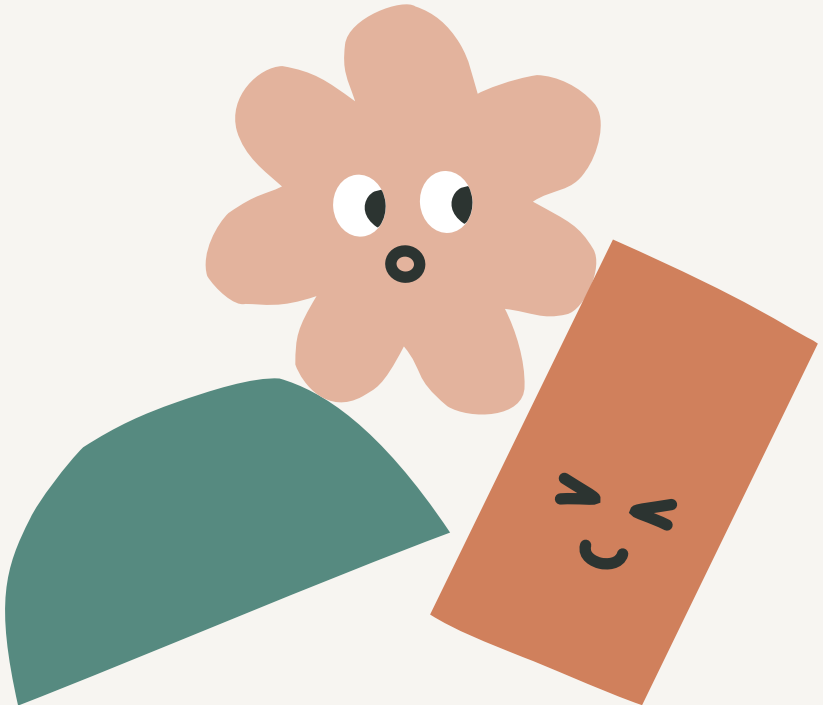
```r
# Set seed so the k-means result is reproducible
set.seed(224)

# Run k-means using all four numeric measurements
iris_kmeans <- kmeans(
  iris[, 1:4],
  centers = 2,
  nstart = 25
)

# Add fitted cluster assignments to the data
iris$kmeans_cluster <- factor(iris_kmeans$cluster)

# Plot fitted clusters (color) against true species (shape)
ggplot(
  iris,
  aes(
    x = Sepal.Width,
    y = Sepal.Length,
    color = kmeans_cluster,
    shape = Species
  )
) +
geom_point(size = 3) +
scale_shape_manual(
  values = c(
    "setosa" = 16,      # circle
    "versicolor" = 17,  # triangle
    "virginica" = 15    # square
  )
) +
labs(
  x = "Sepal Width",
  y = "Sepal Length",
  color = "K-Means Cluster",
  shape = "True Species",
  title = "K-Means Clustering of Iris Data (K = 2)"
) +
theme_minimal()
```

Common AI Mistakes



- Making assumptions that the user did not intend when the prompt was overly broad or was not specific enough
  - Example: User says, “write code to remove all duplicate patients” as part of a longer prompt, AI assumes that it should keep the first entry whenever it finds a duplicate, but actually the most recent entry should be kept
- Lack of structure / organization / comprehensibility
  - AI will often prioritize writing minimal lines over readability
  - AI will write complicated catches / safeguards for irrelevant edge cases or for errors that should break your code
- AI often can’t validate results or ensure that outputs are sensible
- AI will leave mistakes / obsolete code, comments, etc, when editing
- AI does not understand the “why” of your code, it is simply trying to get from point A to point B based on your prompt – **you need to understand exactly what you want from your project and why**
  - Example: User wants to study effect of a pollutant on blood pressure and says to AI “write code to predict blood pressure in this data using pollution”, AI writes code to run the correct regression, but both AI and user failed to think about covariates and confounding leading to incorrect conclusions
- Failing to attribute / explain AI usage for a project, document, etc

Conceptual Understanding

Conceptual Understanding

- Papers are often much easier to understand with the help of AI, it is easy to ask it questions about details or concepts you are confused about, and it is very good at giving a high-level explanation
    - Sometimes AI explanations can gloss over important details when material is complex or lengthy
    - If AI does not itself understand what is going on in a paper, it will simply regurgitate what it read
- Most common statistical concepts you will encounter in coursework are heavily studied and written about on the internet or other sources that AI uses as training data
    - AI can provide useful explanations of coursework concepts and can usually answer detailed questions



Academic Writing

Academic Writing

- Artificial Intelligence has seen an enormous uptick in usage in academic writing – AI it is typically clear and gives well-structured sentences, especially when prompted correctly and given all relevant information
- However, it very often includes a lot of unnecessary words, sentences, and other strange-sounding fluff, so AI generated academic writing should be carefully **read and edited** before sharing / using
- Examples:
- Multiple imputation via Multiple Imputation by Chained Equations (MICE) was employed to handle ~~the~~ missing data ~~that inevitably accumulates over 15 years of longitudinal follow-up~~.
- The study was ~~robustly~~ powered to detect a clinically significant difference **(Of?)** between the control and treatment group's measured CPT performance with power = 0.79.
- The pre-imputation sample exhibited ~~meaningfully~~ missing data across multiple outcomes.
- A ~~notable methodological~~ strength of the present analysis is the use of MICE to handle missing data across both outcomes and covariates, combined with ~~advantageously leveraging~~ **the use of** a longitudinal mixed-effects model ~~framework~~ that exploits the ~~full~~ repeated-measures structure of the SCDS data.

Mathematical Work

Mathematical Work

- My opinions here might be controversial so take with a grain of salt
- I think that modern AI can probably do most / all of the homework and coursework that is required to pass the program
  - If using, be sure to check every logical step in a proof / solution before submitting
  - AI will gloss over crucial steps, and will pretend it has solved a problem it actually has not
  - YOU must understand every step of every solution you turn in, must be able to EXPLAIN and VERIFY
- Your performance in the program is not asocial - you must communicate your understanding
- Demonstrate that you are actively engaged and learning the material, not just trying to do as little as possible by using AI

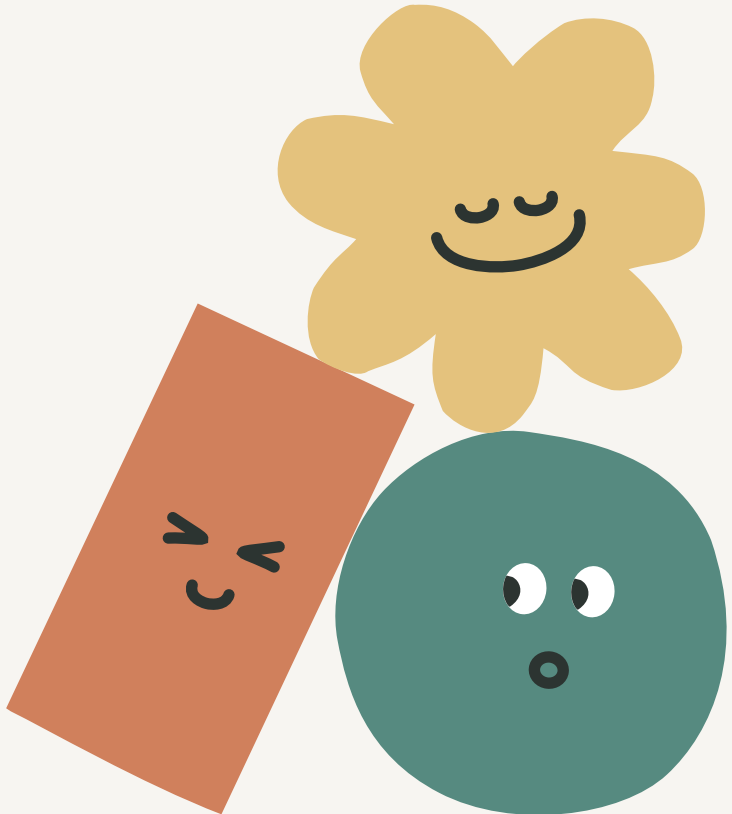


Presentations

“AI doesn’t lift the fog. It raises the cost of not having judgment.”

“Technology doesn’t replace reasoning. It makes the absence of it more expensive.”

– From a presentation about ethical AI usage that was 100% written by Chat GPT



AI Made Presentations



AI Generated Robot

- Just don't do it
- Please don't do it
- It's often easy to tell when a presentation was made by AI, and it reflects poorly on the presenter
- The purpose of a presentation is social: for YOU to communicate YOUR information to the audience in the way that YOU understand it, an AI made presentation turns you into a middle-man, you are now trying to communicate what AI has to say to the audience as AI understands it
- If you are asked a question about something in your presentation it is much easier to explain if YOU wrote what is being asked about and can explain what YOU were thinking or where YOU got whatever it is that is being asked about
- What AI Is good for in presentations:
- Writing individual sentences in the best way
- Organizing the space in a slide in latex / formatting variety, bulleting
- Extremely helpful for **writing latex equations** – just type in English as best as you can or even just take a picture of an equation written in pencil and paper - AI can usually write it out in latex so you don't have to
- AI generated the little space-filler robots-on-laptops and light-bulbs-with-eyes seen throughout this presentation

Conclusions & Future Thoughts

Will AI Replace Us?



- No
- AI is fundamentally changing the way that statistical work is done
- Entering the workforce one's role may be more akin to a manager of statisticians than an entry level statistician since much of the more direct statistical work can be done more efficiently and easily with AI than was ever possible before
- Our primary function in the future (according to me so grain of salt) is going to be:
  - More social and more integrated with collaborators and other stakeholders, verifying, explaining, and ensuring that statistical principles are correctly applied
  - Deciding the directions of greatest importance for statistical work and using judgment / reasonability to decide how best to think about and address statistical problems
  - Driving change in the way that statistical methodology is implemented and the way that statistics related disciplines are conducted
  - Having good answers to the **why** questions of statistical practice and being able to explain and discuss those answers articulately
- AI is an incredibly powerful tool and a lot of effort is wasted by ignoring it
- We need to be extremely **intentional** about how we use it and what it is doing at every step of any sufficiently complex task that we give it
- Thanks for listening!

Thanks!